

Basic Mathematics for Natural Science, Test I (10%)

Part I: Choose the correct answer

- E 1. Let $S = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, then which of the following is wrong?
- a) $\emptyset \in S$ d) $\{\emptyset, \{\emptyset\}\} \subseteq S$
b) $\{\emptyset\} \in S$ e) None
c) $\emptyset \subseteq S$
- D 2. Which statement is the contrapositive of the conditional statement below?
"If tomorrow is Monday, then today is Sunday."
- a) If tomorrow is Sunday, then today is not Monday.
b) If today is Sunday, then tomorrow is Monday.
c) If tomorrow is not Monday, then today is not Sunday.
d) If today is not Sunday, then tomorrow is not Monday. e) None

Part II: Short answer item: write the simplified answer on the space provided

1. If $\neg[\neg r \Rightarrow \neg(p \wedge q)]$ is true, then the truth value of $[(p \Leftrightarrow r) \vee q] \Leftrightarrow (\neg p \wedge r)$ is False.
2. Find the truth values of the following where $U = \mathbb{R}$
 - a) $(\exists x)(\forall y)(x^2 < y^2)$. False.
 - b) $(\forall x)[x \neq 0 \Rightarrow (\exists y)(xy = 4)]$. True.
3. The negation of $p \Rightarrow q$ is $p \wedge \neg q$. True.
4. Let $U = \{x \in \mathbb{Z} : -12 \leq x \leq 6\}$, $C = \{x \in \mathbb{Z} : x = \frac{n}{3}, n \in A\}$, $B = \{0, 1, 3\}$ and $A = \{x \in \mathbb{Z} : x = 2n \wedge x = 3m, \text{ for } -4 \leq n \leq 8 \text{ and } -2 \leq m \leq 3\}$. Then find
 - a. $A \setminus (B \cup C)^c = \{0\}$
 - b. $(A \setminus B) \setminus (A \cup C)^c = \{-6, 6\}$

Part iii: Workout problems; show all the necessary steps

1. Determine whether the following statements are valid or invalid.

If I study, then I will not fail Math 1007.

If I do not play cards to often, then I will study.

I failed Math 1007.

Therefore, I played cards too often.

III) Answer

- ⊗ Assign small letter symbols for the statements.

$P = I \text{ study.}$

$q = I \text{ fail Math1007.}$

$r = I \text{ play cards too often}$

- ⊗ Assign capital letter symbols for the compound statements.

$P: P \Rightarrow \neg q$ ✓

$Q: \neg r \Rightarrow p$ ✓

$R: q$ ✓

$S: r$ ✓

- ⊗ Assume the premises (P, Q & R) are true.

↳ From statement R, $\boxed{q = T}$ ✓

↳ From statement P, $P \Rightarrow \neg q = T$

$$P \Rightarrow \neg T = T$$

$$P \Rightarrow F = T$$

$$\boxed{P = F}$$
 ✓

↳ From statement Q, $\neg r \Rightarrow p = T$

$$\neg r \Rightarrow F = T$$

$$\neg r = F$$

$$\boxed{r = T}$$
 ✓

↳ Since $r = T$, $\boxed{S = \text{True}}$

↳ Therefore, the argument is valid.